

TIC 437022904

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	0.715504 d (ALIASED, see below)	VSX	no match
-> corrected period:	1.431008 d	ASAS-SN	no match
Epoch (t0)	3235.9849 BTJD	Gaia eclipsing-binary table	no match
Duration	0.2440 d	Gaia variable classification	no match
Depth	0.485 % (4,853 ppm)	TESS EB catalog (Prsa+2022)	no match
SNR	573.0	ExoFOP TOI list	no match
INDEPENDENT CHECKS (f3i gates)		Verdict: NOVEL	
Eclipses observed	64 (gate >= 3)		
Robust depth ratio	0.81 (gate >= 0.50)		
Robust depth sigma	75.8 (gate >= 7.0)		
Odd/even z-score	7.00 (ALIAS FLAGGED)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

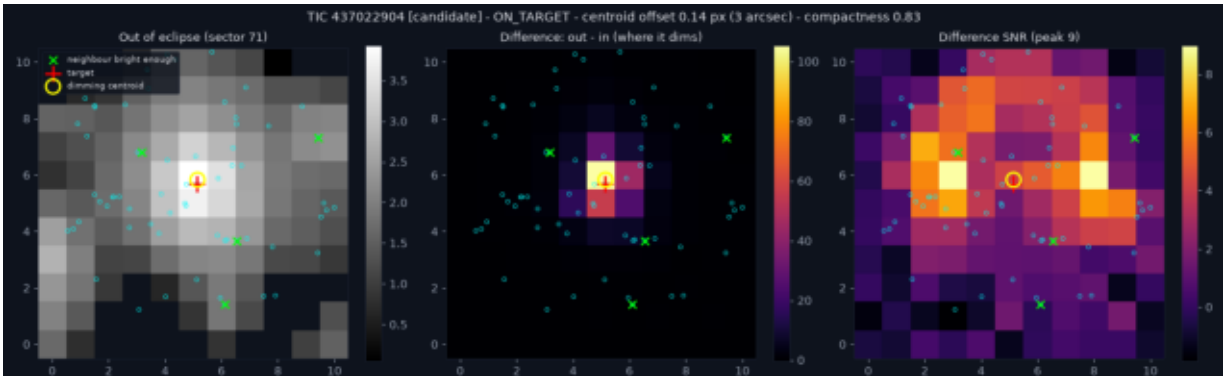
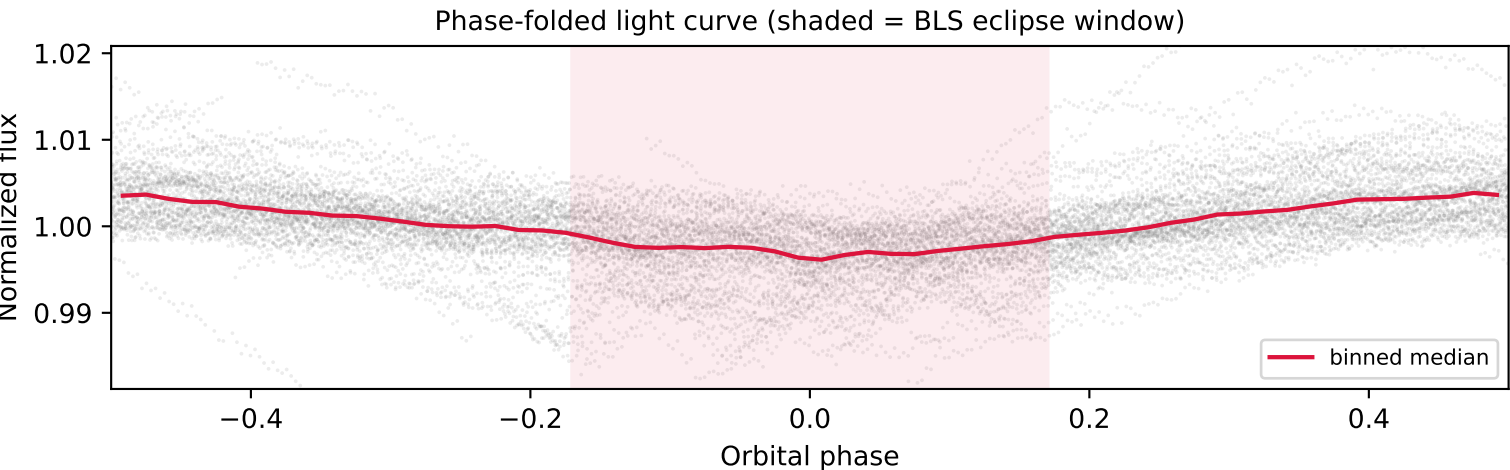
Sector 71 light-curve dip in this sector: 45.0 sigma t0 refined by -0.0475 phase

Difference-image peak SNR 8.99 (gate >= 5.0) Compactness 0.83 (gate >= 0.50)

Dimming centroid offset 0.14 px (3 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 437022914

2.21 px away, dTmag +5.67 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 7.00 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 1.431008 d, in any write-up.